EDF guarantee test (2/2) (Liu & Layland, 1973)

Proof of sufficiency (by contradiction).

- We assume that the task set is not schedulable by EDF and $U < 1$:
  - t_2 := first time instant at which a deadline is missed ∃ PERCHÉ T NON È SCHEDULABILE.
  - $[t_1, t_2]$:= longest interval of continuous utilization before t_2 during which only jobs $\tau_{i,k}$ with arrival time $a_{i,k} \geq t_1$ and absolute deadline $d_{i,k} \leq t_2$ are executed
  - $C_p(t_1, t_2)$:= total processor demand during the interval $[t_1, t_2] \Rightarrow$

JOB RILASCIATI DOPO t_1 E LA CUI DEADLINE È PRIMA DI t_2

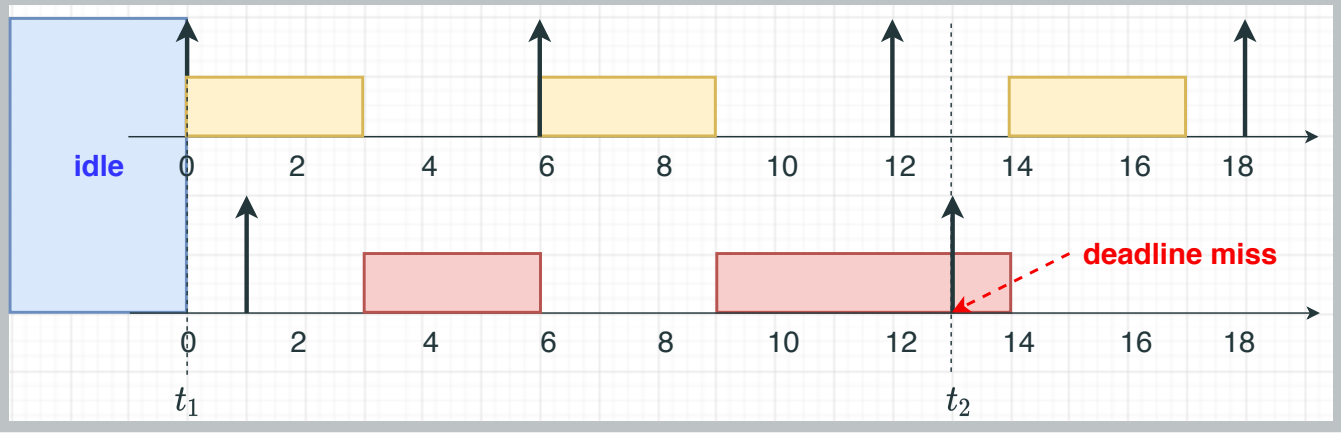
PERCHÉ QUELLI CHE HANNO d_i DOPO t_2 NON MI INTERESSANO

$U_i = \frac{C_i}{T_i}$

$$C_p(t_1, t_2) = \sum_{\tau_{i,k} \mid a_{i,k} \geq t_1 \wedge d_{i,k} \leq t_2} C_i = \sum_{i=1}^{n} \left\lfloor \frac{t_2 - t_1}{T_i} \right\rfloor C_i \leq \sum_{i=1}^{n} \frac{t_2 - t_1}{T_i} C_i = (t_2 - t_1)U$$

#JOB DEL TASK i RILASCIATI IN (t_1, t_2)

- $C_p(t_1, t_2) > t_2 - t_1$ given that a deadline is missed at $t_2 \Rightarrow$
  $\Rightarrow t_2 - t_1 < C_p(t_1, t_2) \leq (t_2 - t_1)U \Rightarrow U > 1$, which is a contradiction



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